

PRINCE SAMUEL KYEREMANTENG

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RESEARCH PROFILE

Computer scientist specializing in software engineering and machine learning systems, with research interests at the intersection of artificial intelligence, software development, and software security. My research focuses on designing intelligent software systems and developer-support tools that incorporate AI capabilities while ensuring reliability, security, and maintainability. Specifically, I am interested in how machine learning components can be systematically engineered, evaluated, and secured within large-scale software systems. My current MPhil research investigates hybrid deep learning architectures for missing-data imputation in IoT sensor networks, combining sequential neural networks (LSTMs) with probabilistic generative models (Variational Autoencoders) to reconstruct incomplete time-series data. Beyond machine learning models, my broader research interests include engineering AI-enabled software systems, such as the secure integration of AI components into software, developer tools that facilitate intelligent programming workflows, empirical evaluation of AI-driven development practices, and software architectures for data-intensive and distributed systems. I am particularly interested in empirical and experimental research methods to understand better and improve the integration of AI technologies into modern software engineering.

RESEARCH INTERESTS

Primary Research Areas

- AI-Enabled Software Engineering (AI4SE, SE4AI, etc.)
- Software Security and Secure Software Development
- Machine Learning Systems Engineering (MLOps)
- Empirical Software Engineering

Specific Topics

- Integration of machine learning components into large-scale software systems
- Security and robustness of AI-enabled software applications
- Developer-support tools for intelligent programming environments
- AI-assisted software development workflows
- Architecture and reliability of data-intensive systems
- Software engineering challenges in IoT-based systems

EDUCATION

Master of Philosophy (MPhil) in Computer Science (In Progress)

University of Ghana - Accra, Ghana

Expected completion: 2026

CGPA: 3.75 / 4.0

Thesis

A Hybrid Deep Learning Framework for Missing Data Imputation in IoT Sensor Data Using LSTM and Variational Autoencoders

Research Contributions

- Designing a hybrid architecture integrating recurrent neural networks (LSTM) with Variational Autoencoders (VAE) for imputing missing observations in multivariate time-series data.

- Investigating the effect of different missingness mechanisms (MCAR, MAR, MNAR) on imputation performance.
- Developing experimental pipelines for evaluating reconstruction accuracy using metrics such as RMSE, MAE, and MAPE.

Application Domain: Environmental monitoring systems and IoT sensor networks.

Bachelor of Science in Computer Science

University of Ghana — Accra, Ghana

2019 – 2023

CGPA: 3.45 / 4.0

Final Year Project

SmartFlow: IoT-Based Smart Irrigation and Environmental Monitoring System

Key Contributions

- Designed a distributed sensor network for real-time monitoring of soil moisture and environmental conditions.
- Implemented automated irrigation control based on real-time sensor analytics.

PUBLICATIONS

Book Chapter

Kyeremanteng, P. S., et al. (2025)

Enhancing Privacy and Efficiency in Distributed AI Systems

In: Edge Computing – Latest Advancements, Challenges, and Applications

Publisher: IntechOpen

DOI: 10.5772/intechopen.1011986

Contributions

- Conducted a literature review on federated learning systems deployed on edge infrastructure.
- Designed architecture models for distributed training pipelines in privacy-sensitive environments.

MANUSCRIPTS UNDER REVIEW

Kyeremanteng, P. S., et al.

AP2-AODV: A Predictive and Context-Aware Probabilistic Route Discovery Mechanism for Dynamic Wireless Networks

Submitted to: Concurrency and Computation: Practice and Experience

Contributions

- Developed probabilistic route discovery model for ad-hoc wireless networks.
- Conducted simulation experiments evaluating routing efficiency and network latency.

WORKING PAPERS

(1) Missing Data Imputation Techniques: Applications, Challenges, and Innovations - A Review

Target journal: International Journal of Information Technology & Decision Making

Focus

- Comparative taxonomy of statistical, machine learning, and deep learning imputation techniques.
- Analysis of domain-specific applications, including healthcare and environmental sensing.

(2) SWIFT: A Parallel K-Mer Filtered Smith-Waterman Extension for Efficient Local Sequence Alignment

Target journal: International Journal of Information Technology & Decision Making

Contribution

- Proposed extension of the Smith-Waterman algorithm using k-mer filtering and tiled dynamic programming.
- Implemented parallelized alignment computation for improved performance on large genomic sequences.

TEACHING EXPERIENCE

Graduate Teaching and Research Assistant

University of Ghana

Department of Computer Science

2023 - 2026

Courses

Artificial Intelligence

Advanced Database Systems

Responsibilities

- Conducted tutorials on machine learning concepts, heuristic search, and knowledge representation.
- Assisted in teaching database transaction management and system design.

RESEARCH SOFTWARE AND TOOLS (DEVELOPED BY ME)

- (1) **ReviewMate:** ReviewMate is a Microsoft Word add-in designed to support academic writing and research workflows using large language models (LLMs).

Core Capabilities

- Automated manuscript review providing structural and linguistic feedback on academic text
- Summarization of research sections to assist literature review and abstract generation
- Paraphrasing and rewriting for improved clarity and conciseness
- Text compression and expansion for refining research arguments

- (2) **ForgeIT:** ForgeIT is a platform designed for generating synthetic datasets for machine learning experimentation, testing, and benchmarking.

Core Capabilities

- Generation of structured synthetic datasets for machine learning experiments
- Configurable data distribution parameters for simulating real-world data variability
- Support for generating missingness patterns (MCAR, MAR, MNAR) for studying missing data imputation methods

INDUSTRY EXPERIENCE

Lead Software Engineer / Consultant

AfriLearnTech Company Ltd. (2025 – Present)

- Leading development of a national digital learning platform for Liberia's Ministry of Education.
- Designing scalable backend systems and distributed software infrastructure.

Founder and Lead Consultant

PKay Software Consultancy (2023 – Present)

- Designed enterprise platforms, including digital admissions systems and business applications.

Software Developer

Payhub Ghana Limited (2021 – 2025)

- Developed USSD-based financial transaction system supporting mobile payments.
- Reduced system latency through Redis-based caching and optimized session management.

GRANTS AND SCHOLARSHIPS

Erasmus+ Mobility Grant

Exchange Program between the University of Ghana and Lund University

January 2026 - June 2026

Funding Amount: €5,628

Purpose: Graduate research mobility and academic collaboration.

GETFund Local Postgraduate Scholarship

For Bright and Deserving Master's Students

Retroactive 2024 – 2026

Scholarship Coverage: Full Tuition Fees for the 2-year duration of the master's program

Ghana Scholarship Secretariat Local Tertiary Scholarship – District Level

For Bright and Deserving Undergraduate Students

2020/21 to 2021/22 academic years.

Scholarship Coverage: Full tuition for 200 and 300 levels.

CONFERENCES AND PRESENTATIONS

(1) Postgraduate Research Workshop

University of Ghana (2025)

Presentation

Missing Data Imputation Techniques: Applications, Challenges, and Innovations

Award

3rd Best Paper Award

(2) Nyansapo AI Symposium on Sustainable Development

Organized by the Ghana Atomic Energy Commission and Université Paris-Saclay (2023)

Presentation: AI-based predictive system for electricity outage forecasting.

TECHNICAL SKILLS

Programming	Machine Learning	Frameworks	Systems
Python	PyTorch	Django	Git
JavaScript	TensorFlow	Flutter	Linux
SQL	ScikitLearn	FastAPI	Redis
Dart		React	Docker

ACADEMIC SERVICE AND LEADERSHIP

Student Mentor

Department of Computer Science

University of Ghana (2024 – 2026)

Mentoring undergraduate students in programming, software development, and project design.

REFEREES

Available upon request.